

College Mini-Project Demonstration Document

Full-Stack File Search and Streaming Download Web System

1. Project Objective & Implementation Overview

This web application demonstrates a secure, high-performance file search and retrieval engine.

The frontend is built with React.js featuring real-time search, responsive cards, and theme toggling.

The backend is built with Node.js and Express.js, providing REST APIs on port 5000.

Files are stored locally in the backend files directory, without external database dependencies.

2. Search and Download Architecture

API Search Endpoint: GET /api/files/search?q=<query> performs case-insensitive substring search.

API Download Endpoint: GET /api/files/download/<filename> securely streams binary file attachments.

Security Guard: Blocks directory traversal (e.g., ../ attacks) using path boundary verification.

3. Verification & Evaluation Checklist

[PASSED] Case-insensitive partial filename matching verified (e.g. searching "math").

[PASSED] Secure file streaming download via Express res.download().

[PASSED] Graceful handling of empty results, loading states, and backend disconnection.